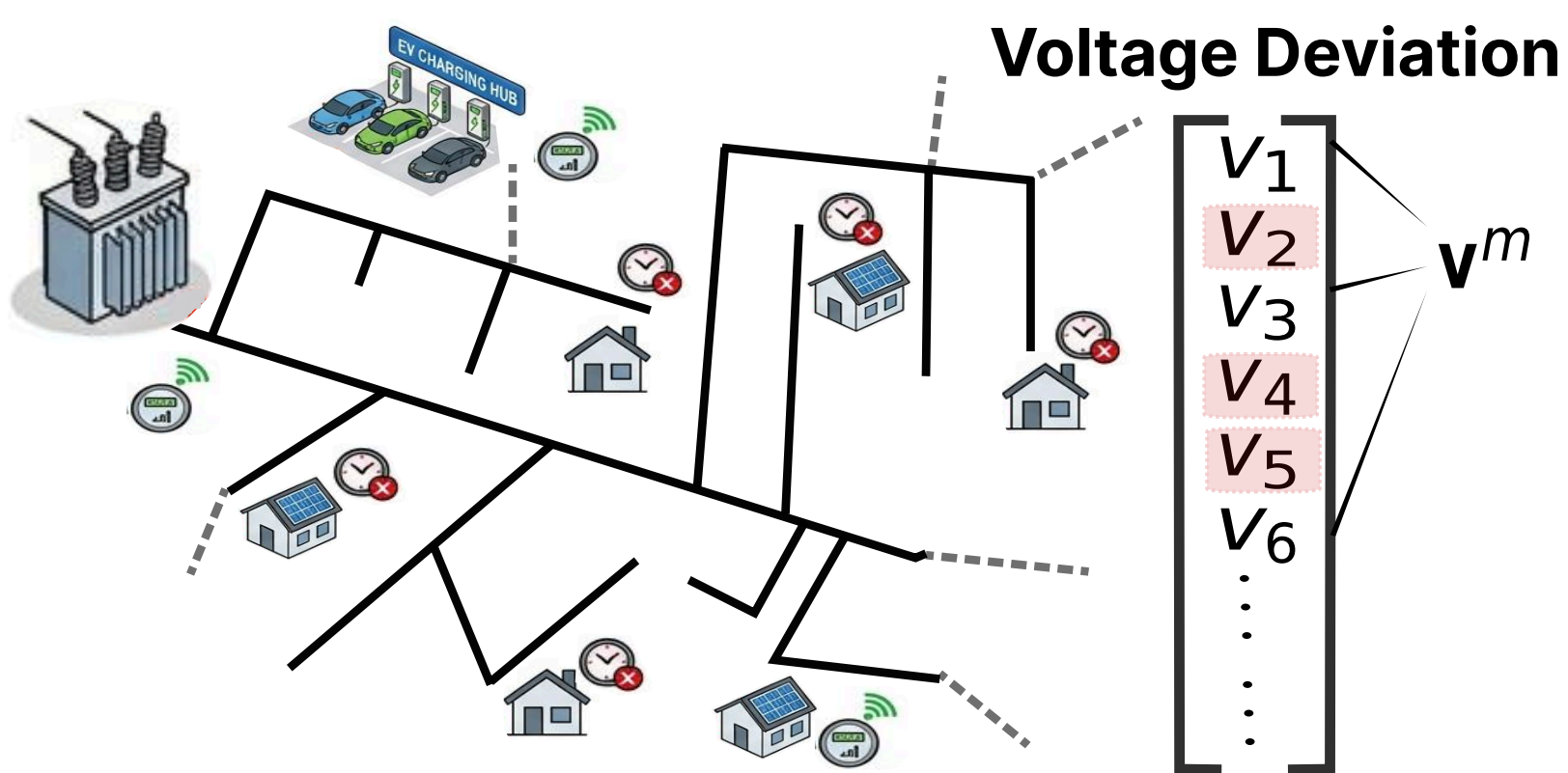


Optimal Measurement Selection for Certifiable Voltage Monitoring in Distribution Grids

Peng Zhang, Line Roald, Manish Singh

Motivation

Voltage Monitoring: use available measurements to assess the compliance of prescribed voltage safety limits $[\mathbf{v}^{\min}, \mathbf{v}^{\max}]$.



Limited Bandwidth
(Low Reporting Rate)

Partial Real-Time
Measurement

- Pseudo-measurements result in non-certifiable safety at unobserved nodes.
- Different measurement selections provide distinct safety perception.

Research Questions

Worst-case achievable voltages with bounded power injections [R1], [R2]:

$$f_i(\mathbf{v}^m) = \min_{\mathbf{x}} v_i \quad \text{or} \quad g_i(\mathbf{v}^m) = \max_{\mathbf{x}} v_i \quad (i \in \mathcal{N})$$

$$\text{s.t. } \mathbf{x} \leq \mathbf{x} \leq \bar{\mathbf{x}},$$

$$\mathbf{C}^m \mathbf{x} = \mathbf{v}^m,$$

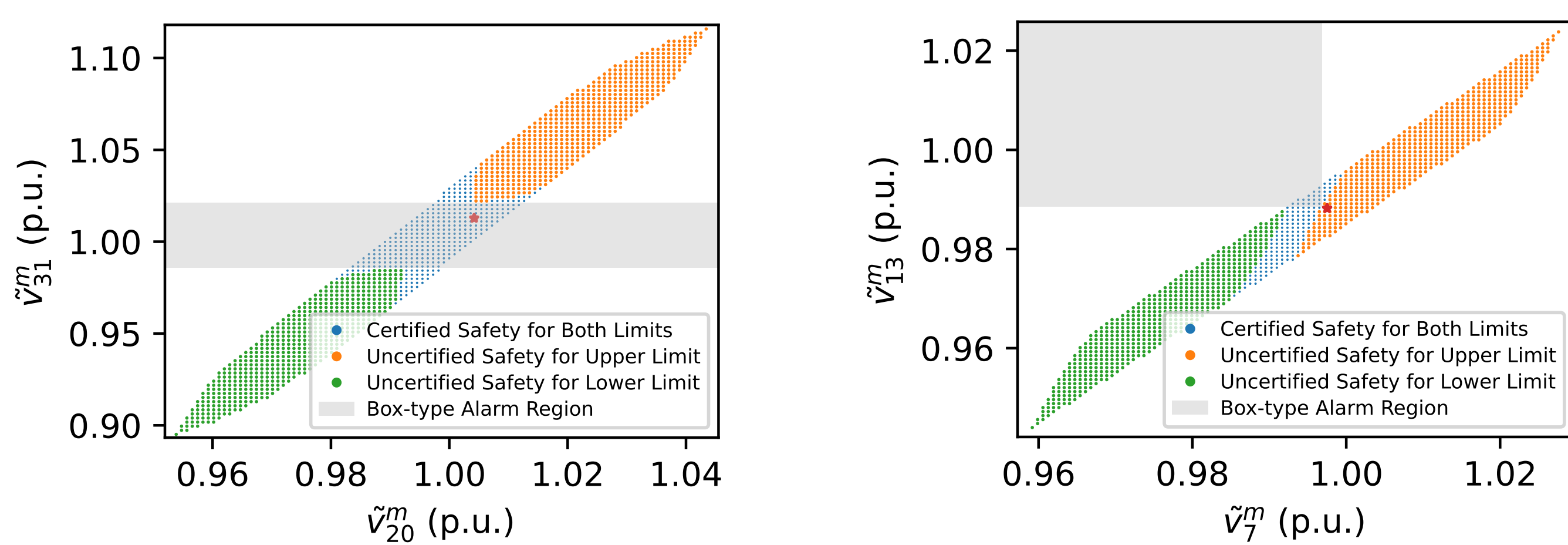
$$\mathbf{c}_i^\top \mathbf{x} = v_i.$$

$\mathbf{x}$: power injection
($\mathbf{C}^m, \mathbf{c}_i$): LinDistFlow model

ASK: How to select the sites $\mathcal{N}^m$ for real-time voltage measurement reporting $\mathbf{v}^m$ and safety certification?

Problem Formulation

Geometric Insights



Metric to Optimize: Aggregated Violation of Safety Certification

$$H(\mathcal{N}^m; \mathbf{v}^m) = \sum_{t=1}^T \sum_{i=1}^N [v_i^{\min} - \tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t)]_+ + [g_i(\mathbf{v}_t, \mathbf{v}_t) - v_i^{\max}]_+$$

Bilevel Problem for Optimal Measurement Selection

Given full voltage profile $\mathcal{D} = \{\mathbf{v}_t\}_{t=1}^T$ and budget ρ :

$$\begin{aligned} \min \quad & \bar{H}(\mathbf{s}; \mathcal{D}) \\ \text{over } \quad & \mathbf{s} \in \{0, 1\}^N, \mathbf{v}_t, \mathbf{v}_t \in \mathbb{R}^N, \forall t \\ \text{s.t. } \quad & \mathbf{1}_N^\top \mathbf{s} \leq \rho, \\ & \bar{H}(\mathbf{s}; \mathcal{D}) = \sum_{t=1}^T \sum_{i=1}^N ([v_i^{\min} - \tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t)]_+ + [g_i(\mathbf{v}_t, \mathbf{v}_t) - v_i^{\max}]_+), \\ & \mathbf{v}_t = \text{Diag}(\mathbf{s})\mathbf{v}_t - M(\mathbf{1}_N - \mathbf{s}), \\ & \mathbf{v}_t = \text{Diag}(\mathbf{s})\mathbf{v}_t + M(\mathbf{1}_N - \mathbf{s}), \\ & \tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t) = \min_{\mathbf{x}} \mathbf{c}_i^\top \mathbf{x}, \text{ s.t. } \mathbf{x} \in [\mathbf{x}, \bar{\mathbf{x}}], \mathbf{C}\mathbf{x} \in [\mathbf{v}_t, \bar{\mathbf{v}}_t], \\ & \tilde{g}_i(\mathbf{v}_t, \mathbf{v}_t) = \max_{\mathbf{x}} \mathbf{c}_i^\top \mathbf{x}, \text{ s.t. } \mathbf{x} \in [\mathbf{x}, \bar{\mathbf{x}}], \mathbf{C}\mathbf{x} \in [\mathbf{v}_t, \bar{\mathbf{v}}_t] \end{aligned}$$

**Big-M
Formulation**

Methods

Single-level Reformulation

• Epigraph: $\min [v_i^{\min} - \tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t)]_+ \Rightarrow \min l_{i,t}, \text{ s.t. } v_i^{\min} - l_{i,t} \leq \tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t).$

• Strong-duality of $\tilde{f}_i(\mathbf{v}_t, \mathbf{v}_t)$:

$$\begin{aligned} v_i^{\min} - l_{i,t} &\leq \alpha_{i,t}^\top \mathbf{x} - \alpha_{i,t}^\top \mathbf{x} + \beta_{i,t}^\top \mathbf{v}_t - \beta_{i,t}^\top \mathbf{v}_t, \\ \mathbf{c}_i - \alpha_{i,t} + \bar{\alpha}_{i,t} - \mathbf{C}^\top (\beta_{i,t} - \bar{\beta}_{i,t}) &= 0, \\ \alpha_{i,t}, \bar{\alpha}_{i,t}, \beta_{i,t}, \bar{\beta}_{i,t}, l_{i,t} &\geq 0. \end{aligned}$$

$2 \times N \times T$
Bilinear Constraints

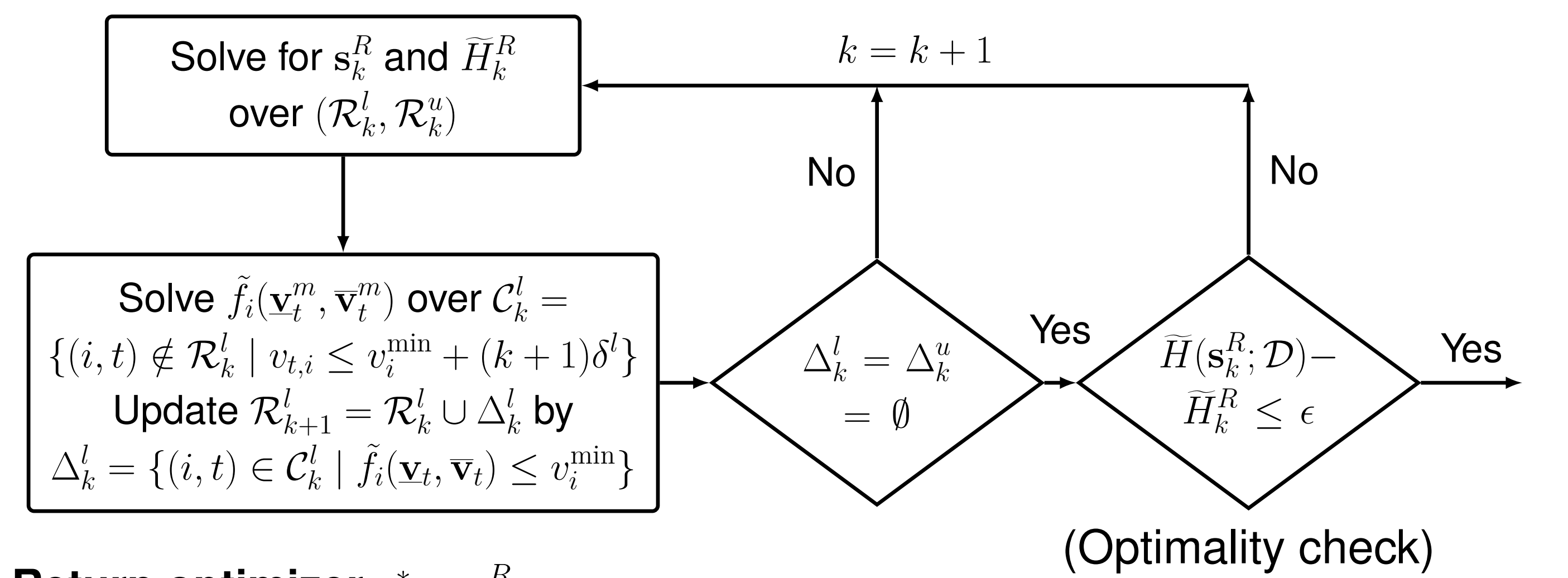
Monotonicity of $\bar{H}(\mathbf{s}; \mathcal{D})$ (Prop. 3.1)

$\bar{H}(\mathbf{s}_1; \mathcal{D}) \geq \bar{H}(\mathbf{s}_2; \mathcal{D})$, if $\mathbf{s}_1 \leq \mathbf{s}_2$ (nested structure).

$$\mathbf{1}_N^\top \mathbf{s} \leq \rho \longrightarrow \mathbf{1}_N^\top \mathbf{s} = \rho$$

Scenario Reduction (Algorithm 1)

Initialize $\mathcal{R}_0^l = \{(i, t) \mid v_{t,i} \leq v_i^{\min}\}$ and $\mathcal{R}_0^u = \{(i, t) \mid v_{t,i} \geq v_i^{\max}\}$,



Return optimizer $\mathbf{s}^* = \mathbf{s}_k^R$.

(Optimality check)

Results

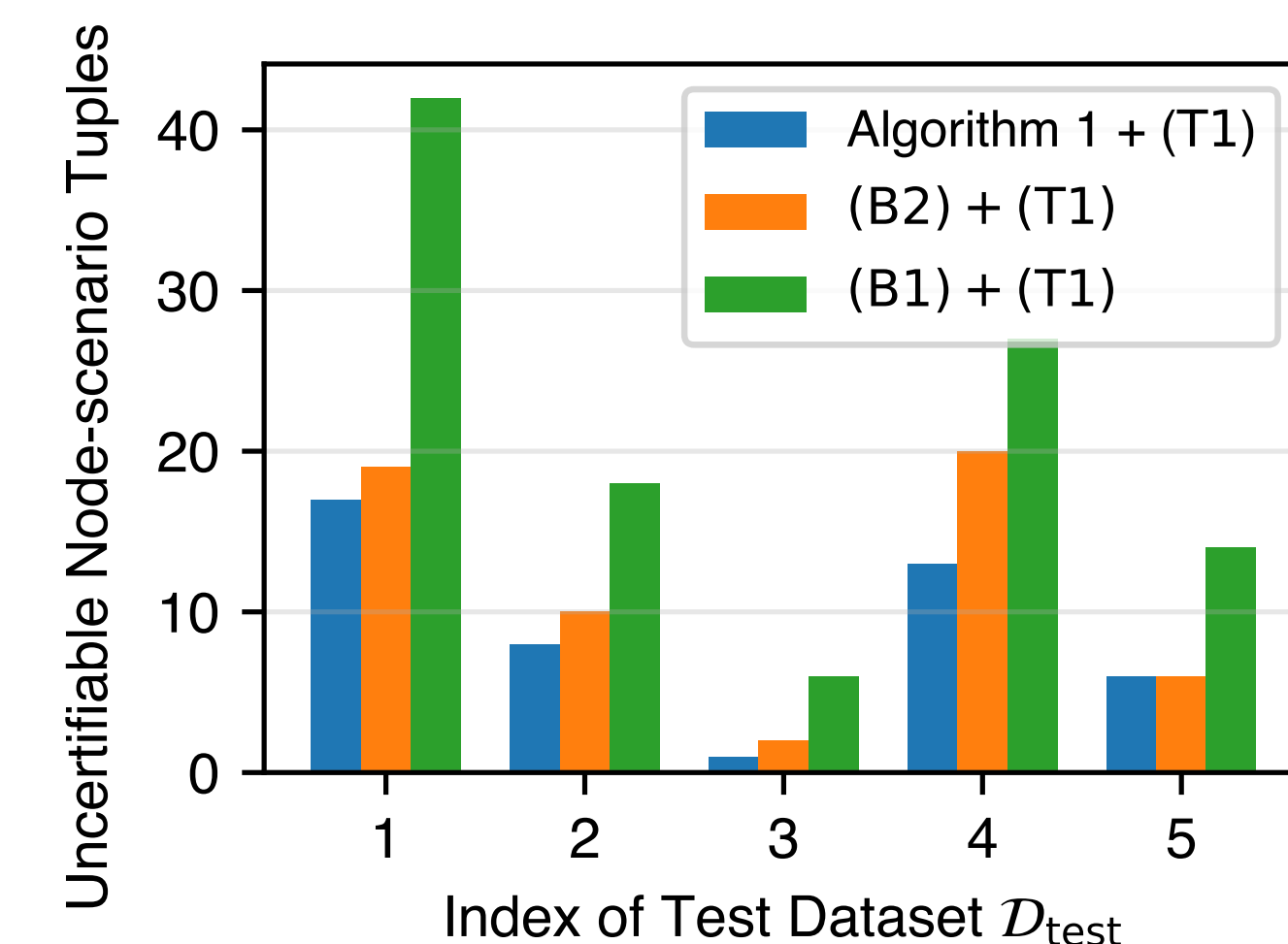
Setup: SCE 56-bus feeder with power variability $\pm 60\%$ (load) and $\pm 100\%$ (PV); Datasets $\mathcal{D}$ with $T = 288$ (5-min intervals over a day); Budget $\rho = 8$.

Solving Capability (Averaged over 10 datasets)

	$\tilde{H}^R [\times 10^{-2}]$	Gap	$ \mathcal{R}^l + \mathcal{R}^u $	Time [s]
Iterative scenario reduction	3.3	$< 10^{-4}$	21	43.6
Full single-level problem	—	—	$2 \times 55 \times 288$	900

(Time for optimality check ≈ 4 min)

Performance of Selected Measurements



- B1: nodes with highest and lowest voltages
 - B2: nodes with largest size of box-type alarm region [R1]
- Reduction on safety-uncertified measurements 58% and 21%**

Conclusion

- Bilevel problem for measurement selection with a novel violation metric
- Single-level reformulation and iterative scenario reduction algorithm

